Code-Layout, Readability and Reusability

SMARTBRIDGE - Code-Layout, Readability and Reusability

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-7817
- **Team Size:** 4
- **Team Leader:** Karan S
- **Team Members:** SANJAY .M, Santhosh S, Yaseen P
- **Project Name:** AI FitTrack
- **Maximum Marks:** 5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	**Consistent Indentation**	Uniform spacing/tabs used throughout the code	**Yes**	Standardized 2-space JavaScript formatting enforced across all controllers and services.
2	**Proper File Structure**	Files and folders are logically organized	**Yes**	Modular MVC layout (src/config, controllers, middleware, models, routes, services).
3	**Meaningful Variable Names**	Variables reflect their purpose clearly	**Yes**	Descriptive variable identifiers used (e.g., workoutName, caloriesBurned, averageDuration).
4	**Function / Method Names**	Functions are descriptively named	**Yes**	Verb-driven descriptive names (e.g., generateWorkoutRecommendation, matchPassword).

5	**Code Comments**	Inline and block comments explain logic	**Yes**	Detailed block comments for pre-save hooks, JWT checks, and Gemini service calls.
6	**Modular Design**	Code is split into reusable functions/modules	**Yes**	Business logic, authentication middleware, and AI APIs isolated in dedicated modules.
7	**No Redundant Code**	Duplicate or unused code is removed	**Yes**	Dry principle maintained using centralized middleware (auth.js, errorHandler.js).
8	**Error Handling**	Exceptions and errors are handled gracefully	**Yes**	Asynchronous try-catch blocks and centralized JSON error handler responses.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	**connectDB**	Node.js / Mongoose	Database initialization on application startup (server.js)	**High**
2	**protect (Auth Middleware)**	Express.js / JWT	Applied across all secure Workout and AI endpoints (workoutRoutes.js, aiRoutes.js)	**High**

3	**geminiService**	Node.js / @google/genai	Called inside aiController.js for custom workout generation and stats analytics	**High**
4	**matchPassword Hook**	Mongoose / bcryptjs	Reused inside authController.js during user login verification	**High**
5	**errorHandler Middleware**	Express.js	Centralized error formatting layer applied globally across Express routes	**High**

Overall Code Quality Assessment

Aspect PDF	**Rating (1-5) PDF**	**Comments PDF**
Code Layout & Structure	**5 / 5**	Strict MVC separation ensures clean, maintainable folder hierarchy.
Readability	**5 / 5**	Clear naming conventions and consistent formatting throughout all JavaScript files.
Reusability	**5 / 5**	Service-oriented layer isolates Gemini AI integration and JWT authentication cleanly.
Documentation / Comments	**4.5 / 5**	Comprehensive inline comments and clear error logging in service handlers.

Overall Score	**4.9 / 5**	**Excellent enterprise-grade RESTful API backend structure.**